

Static and formal verification

Slide 1 · 0–2 min

Static and formal verification

Every technique must discharge a named verification claim.

Establish the static boundary → Qualify the proof → Choose the application

Teaching script

Open with a decision the audience must make in this area. Ask which result they currently treat as sufficient evidence and keep that answer for the closing discussion.

The lesson selects a practical argument from the book. The study handout preserves the detailed examples, citations and worked artifacts.

Source: ch13:15–55, ch14:19–64, ch15:17–45, ch16:17–65

Learning outcomes

Slide 2 · 2–5 min

Learning outcomes

Demonstrate these decisions on a real plan row.

- Separate structural results from protocol obligations.
- Qualify a proof by assumptions and abstraction.
- Distinguish generated intent from independent checking.
- Combine typed results through a verification plan.

Teaching script

Read the outcomes as observable tasks. Ask learners to identify which one they can already demonstrate and which one currently depends on undocumented judgment.

At closure, require a concrete artifact or explanation for an outcome rather than a general expression of confidence.

Source: ch13:15–55, ch14:19–64, ch15:17–45, ch16:17–65

Route through the lesson

Slide 3 · 5–7 min

Route through the lesson

Four blocks, each ending with a decision.

- Establish the static boundary
- Qualify the proof
- Choose the application
- Combine evidence without flattening it

Teaching script

Allow eighteen minutes for each block: a concept, its engineering consequence, a worked interpretation and a short discussion. Keep the discussion tied to the evidence available in the example.

Reserve the final eleven minutes for the assigned exercise and application to the learners' work. The timing is a course design allocation, not an engineering estimate.

Source: ch13:15–55, ch14:19–64, ch15:17–45, ch16:17–65

Static results have a subject

Slide 4 · 7–11 min

Static results have a subject

Structural cleanliness does not establish functional correctness.

Structure → Rules and waivers → Remaining behavior

Teaching script

Use lint to remove avoidable structural ambiguity before spending simulation or proof effort. Ask which failure classes the selected rules actually detect.

A clean report says nothing about a missing specified feature unless that feature has an independent check. Keep rule configurations and waivers attached to the result.

Source: ch13:15–55, ch13:56–109

Synchronizers do not define a protocol

Slide 5 · 11–16 min

Synchronizers do not define a protocol

Independent bit crossings do not guarantee a coherent word.

- Review the crossing structure.
- Then review data stability, capture and handshake behavior.

Teaching script

Draw a multibit value whose bits can be observed from different source updates. Individually synchronized bits do not make those updates atomic.

The structural CDC result and the protocol obligation need separate evidence. Ask what guarantees the data is stable during the destination's capture window.

Source: ch13:110–151, ch13:152–214

Reset and unknowns need their own review

Slide 6 · 16–21 min

Reset and unknowns need their own review

A clock-domain inventory does not enumerate reset hazards.

- Inspect reset assertion and release interactions.
- Separate X optimism from X pessimism.

Teaching script

Explain why reset interaction can cross boundaries even where clocks are shared. Do not infer RDC closure from a clean CDC report.

An unknown that disappears optimistically and one propagated pessimistically call for different diagnoses. The simulation configuration is part of the interpretation of the result.

Source: ch13:215–260, ch13:261–350

Discussion: a clean crossing report

Slide 7 · 21–25 min

Discussion: a clean crossing report

Name the evidence that would settle the decision.

- A multibit status uses independent synchronizers.
- The structural report is clean.
- What coherent-transfer claim still needs evidence?

Teaching script

Expected answer: the receiver must observe a value that corresponds to a permitted source transaction, not an arbitrary mixture of updates.

Ask for a protocol argument and its checker. Neither the absence of a structural warning nor the name of a synchronization cell supplies that argument.

Source: ch13:110–151, ch13:152–214

A proof verdict is a qualified statement

Slide 8 · 25-29 min

A proof verdict is a qualified statement

Property, model and assumptions travel with the result.

Property → Model and assumptions → Qualified verdict

Teaching script

Have a learner say the complete claim aloud: which property, over which model, under which environment assumptions. A green result without these qualifiers cannot be transferred to a plan row.

Ask which legal behavior an assumption removes. A restriction introduced only to make convergence easier needs independent review before its proof can support sign-off.

Source: ch14:19-64, ch14:65-158

A bounded pass has a boundary

Slide 9 · 29–34 min

A bounded pass has a boundary

State the depth and justify what lies within it.

- A counterexample is a concrete failing trace.
- A bounded pass needs a depth argument.

Teaching script

Contrast finding a failure within a bound with establishing absence of failure beyond it. The latter does not follow automatically.

The chapter combines latency and scenario analysis, cover witnesses and prior counterexamples. Review completeness of the scenario enumeration as well as its arithmetic.

Source: ch14:159–232, ch14:336–460

Abstraction changes the trace set

Slide 10 · 34–39 min

Abstraction changes the trace set

Know which conclusion survives the reduction.

- Adding behavior can preserve proofs but introduce false failures.
- Removing behavior can preserve failures but hide real ones.

Teaching script

Use over-approximation and under-approximation explicitly. Explain why admitting extra traces makes a successful universal proof stronger while a failure may need validation against the concrete design.

Conversely, restricting traces can make a discovered failure useful without making a pass exhaustive. Record deleted behavior and the plan rows that cover it elsewhere.

Source: ch14:233–282

Discussion: assume away the failure

Slide 11 · 39–43 min

Discussion: assume away the failure

Name the evidence that would settle the decision.

- An ordering property proves after adding an assumption.
- No owner can justify that environment restriction.
- Can the row close?

Teaching script

Expected answer: the result does not discharge the original unrestricted claim. Review or withdraw the assumption, then obtain evidence for the remaining legal behavior.

The chapter's assumption register distinguishes proven premises, simulation checks and accepted arguments. An undocumented convergence aid belongs to none of those categories.

Source: ch14:65–158, ch14:336–460

Automation still needs reviewed intent

Slide 12 · 43–47 min

Automation still needs reviewed intent

Generated properties inherit their source description.

- Connectivity and register checks scale repetitive obligations.
- Review the description that generated the checks.

Teaching script

A connectivity specification or register description lets the tool generate many mechanical checks. The engineer still decides whether that description captures the intended connection or side effect.

Generation can eliminate transcription drift while preserving a shared error in the source. Ask which review or independent evidence can expose that shared error.

Source: ch15:17–45, ch15:46–77, ch15:78–111

An inferred property is not an independent oracle

Slide 13 · 47-52 min

An inferred property is not an independent oracle

Mining a design can reproduce its mistake.

- Label the origin of generated properties.
- Compare design-derived claims with external intent.

Teaching script

Distinguish elaborating a reviewed specification from discovering patterns in the implementation. Both may produce useful properties, but they have different independence.

A property that faithfully describes current RTL can hold while the RTL violates its specification. Keep that distinction visible when admitting the property into a closure argument.

Source: ch15:147-160

Equivalence inherits the reference

Slide 14 · 52–57 min

Equivalence inherits the reference

Preservation and correctness are different claims.

- Define the correspondence and transformation boundary.
- Choose sequential reasoning when state alignment changes.

Teaching script

Use a combinational transformation and a retiming change to expose the difference between matched-state comparison and sequential equivalence.

A successful comparison supports preservation within the configured model. It cannot independently establish that the reference was correct, nor that excluded modes behave acceptably.

Source: ch15:161–219

Discussion: generated from one source

Slide 15 · 57–61 min

Discussion: generated from one source

Name the evidence that would settle the decision.

- The register RTL and its checks share a description.
- All generated checks pass.
- Which error can both artifacts share?

Teaching script

Expected answer: an incorrect or omitted behavior in the source description can be reproduced consistently in both artifacts.

Ask for the external specification review or independent check that would expose it. A second generated artifact is not automatically a second interpretation of intent.

Source: ch15:78–111, ch15:147–160

Allocate engines per claim

Slide 16 · 61–65 min

Allocate engines per claim

A feature name alone does not select a method.

- Match the oracle, state space and runtime need.
- Keep each result attached to its plan row.

Teaching script

Take one DMA feature and separate an invariant, a functional transfer and a long software scenario. They can need different engines and evidence boundaries.

The plan should explain why each engine answers the particular row. Do not choose a method merely because a tool or platform is available.

Source: ch16:17–65, ch16:169–199

Check the formal environment in simulation

Slide 17 · 65–70 min

Check the formal environment in simulation

An assumption is a promise that another boundary must honor.

- Turn relevant premises into simulation checks.
- Cover important boundary behavior.

Teaching script

The same expression can restrict a formal environment and monitor the actual integration environment. Explain why these are complementary uses with different verdicts.

A simulation check observes sampled executions; it does not prove the environment universally. Record that scope when discharging the formal assumption.

Source: ch16:66–126

Keep coverage evidence typed

Slide 18 · 70–75 min

Keep coverage evidence typed

Simulation activity and formal coverage have different denominators.

- Unify the plan, not unrelated percentages.
- Handoff assumptions, exclusions and unresolved traces.

Teaching script

Ask what an average of a functional bin percentage and a proof-core percentage would mean. There is no shared population to support that aggregate.

Use the plan row as the common unit. A handoff must retain the exact residual claim and its model restrictions so that the receiving engine does not unknowingly check a weaker problem.

Source: ch16:127–168, ch16:200–285

Discussion: the blended dashboard

Slide 19 · 75–79 min

Discussion: the blended dashboard

Name the evidence that would settle the decision.

- A dashboard averages simulation and formal coverage.
- The resulting percentage rises.
- Which closure argument is still missing?

Teaching script

Expected answer: the mapping from each typed result to the intended feature claim, with its own denominator and limitations.

Require the presenter to name what newly became established. A rising arithmetic aggregate cannot answer that question by itself.

Source: ch16:127–168

Exercise: coverage or mutation sensitivity?

Slide 20 · 79–85 min

Exercise: coverage or mutation sensitivity?

Choose an instrument for the actual question.

- Proof-core coverage is offered as arbitration-change sensitivity.
- Explain the gap and choose the appropriate instrument.
- Include the chapter's recommended proof-core scope.

Teaching script

Exercise 14.4 (judgement). A team offers proof core coverage, run over every property in its suite, as evidence that the property set would notice a change to the crossbar's arbitration logic. Decide what the four coverage kinds of §14.6 establish as a class and what none of them establishes, and name the instrument §14.6 assigns to the question the team is actually asking. State the scoping §14.6 gives for proof core analysis, with the two reasons it gives for that scoping.

Assign book Exercise 14.4 in full. Review the four coverage kinds before the learner selects an instrument.

Expected answer: coverage measures aspects of exploration or the proof's relation to implementation; none establishes that a property set detects a meaningful design mutation. Mutation analysis addresses that question. Explain the chapter's selected proof-core scope and its cost and interpretation limits from the reading. Scope proof-core analysis to a handful of highest-value properties: it has the highest computational cost and its vendor-specific metrics are harder to interpret.

Source: ch14:336–460

What to change on Monday

Slide 21 · 85–87 min

What to change on Monday

Make one existing verification decision reviewable.

- Audit an assumption without a discharge owner.
- Find a crossing with no protocol checker.
- Replace one blended metric with per-claim evidence.

Teaching script

Ask each learner to choose one of these actions in an existing project. Require a small, inspectable deliverable and identify the person who will review it.

Keep the change connected to the lesson's evidence limits. A new dashboard, script or model is useful only if it strengthens the claim the plan row is meant to establish.

Source: ch13:15–55, ch14:19–64, ch15:17–45, ch16:17–65

Scope and next reading

Slide 22 · 87–90 min

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Static and formal verification handout and book Chapters 13–16.
- Tool recipes and complete proof harnesses remain in the handout.

Teaching script

Revisit the outcomes and ask the room to state a limitation of the strongest result discussed today. A complete answer identifies a model boundary, an assumption, a missing oracle or the scope of the underlying evidence.

Assign the detailed reading to the relevant chapters. The omitted topics are explicit scope boundaries, not claims that those topics are unimportant.

Source: ch13:15–55, ch14:19–64, ch15:17–45, ch16:17–65